



## COS110 Strings And Files

Computer science (University of Pretoria)



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COS110 - Program Design: Introduction

Working with Strings & Text Files

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# 1 String Manipulation

The standard string class in C++ has useful built-in functions for working with strings. Ensure that you are familiar with the following functions, since it will help you solve the assignments and practicals in COS110.

Ensure you include the string class to have access to the following functions.

```
#include <string>
```

1

**Declaration:**

```
string str = "Welcome to COS110";
```

1

| Function                   | Explanation                                                                                                                                                    |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| str.length()               | Returns the length of the string str.                                                                                                                          |
| str.at(i) or str[i]        | Used to access the character at position i in str.                                                                                                             |
| str.append(str1)           | Append str1 to the end of str.                                                                                                                                 |
| str.compare(str1)          | Compare str and str1: If the strings are equal, return zero. If str is greater than str1, return value $> 0$ . If str1 is greater than str, return value $< 0$ |
| str.substr(start, offset)  | Extract a substring starting at start and ending at start+offset in str. Therefore offset is the number of characters to be copied.                            |
| str.find(str1)             | Returns the position of the first occurrence of str1 in the str. If str1 is a substring, it will return the position where str1 starts in str.                 |
| str.replace(i, size, str1) | Replace the section of a str that starts at i up to position i+size with str1.                                                                                 |
| str.insert(i, str1)        | Insert a str1 at position i in str.                                                                                                                            |
| str.erase(start, end)      | Erase the section of str that starts at start and ends at end.                                                                                                 |

Table 1: String Manipulation Functions

For more information about C++ strings and C style strings, please have a look at Chapter 9 of the prescribed textbook.

## 2 Text Files

Text files are often used to store data that need to be processed. In COS110 most of the input for your assignments and practicals will need to be read from a text file. For this reason it is crucial that you know how to work with text files.

To work with a text file in your C++ project, you have to include the fstream library:

```
#include <fstream>
```

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### Creating or Opening a text file:

```
ofstream my_file("output.txt"); //Will open the file output.txt if it exists
                                //or create one if it does not.
ofstream my_file("input.txt"); //Will open the file input.txt (can only read
                                //from input.txt)
```

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### Writing to a text file:

```
my_file << "Some_important_text_here!\n";
```

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### Reading from a text file:

```
string str;
my_file >> str; // This will read the line at the current line pointer of my_file
                // and store it in str
```

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To read the lines in a text file one by one until the end of the file is reached you must use a while loop:

```
while (getline (my_file, str)){ //Will be false if all the lines
                                //in my_file was read
    //Do something with str
    cout << str << endl;
}
```

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### Closing a text file:

```
my_file.close();
```

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It is good practice to close text files since it frees up memory. For more information about working with text files, please refer to Chapter 2.4 and Chapter 12 of the prescribed textbook.